

Introduction to Computer Simulation II

Homework 09

Due: Tuesday, 16 June 2026

17. Multiple-histogram reweighting

Write a computer program for multiple-histogram reweighting, the so-called “Weighted Histogram Analysis Method (WHAM)”, and test it for the 2D 16^2 Ising model with the three simulation points $\beta_1 = 0.375$, $\beta_0 = \beta_c = \ln(1 + \sqrt{2})/2 \approx 0.440\,686\,79\dots$, $\beta_2 = 0.475$ (cp. problem 22 of Computer Simulation I).

18. Master plot for the susceptibility

Determine the susceptibility χ of the 2D Ising model on lattices of size 16×16 , 32×32 , 64×64 , and 128×128 for 31 inverse temperatures β in the high-temperature phase in the range $0.85\beta_c, \dots, \beta_c$ close to the critical point at β_c analogously to problem 16. Plot these data

- i) as function of β and then
- ii) in a log-log plot as function of $b \equiv 1 - \beta/\beta_c$. Here it does make sense to compare with $b^{-7/4}$. Finally
- iii) generate the Master plot of $\chi L^{-\gamma/\nu}$ as function of the scaling variable $x = (1 - \beta/\beta_c)L^{1/\nu}$ in a log-log representation. Use for this illustration of the scaling property the exact critical exponents of the 2D Ising model $\nu = 1$ and $\gamma = 7/4$ and compare with $x^{-7/4}$.